

STA 5635 HW 6

Palmer Swanson

February 14, 2026

Problem 1

- (1a) Two random forest classifiers. One with 22 features, the other with 500. Table in figure 1

Model	Train Misclass. Error	Test Misclass. Error
Random Forest, 22 Features	0.000000	0.291667
Random Forest, 500 Features	0.000000	0.165000

Table 1: Train and Test Misclassification Errors, part a

- (1b) Compute feature importances using mean decrease in impurity and the permutation feature importances.

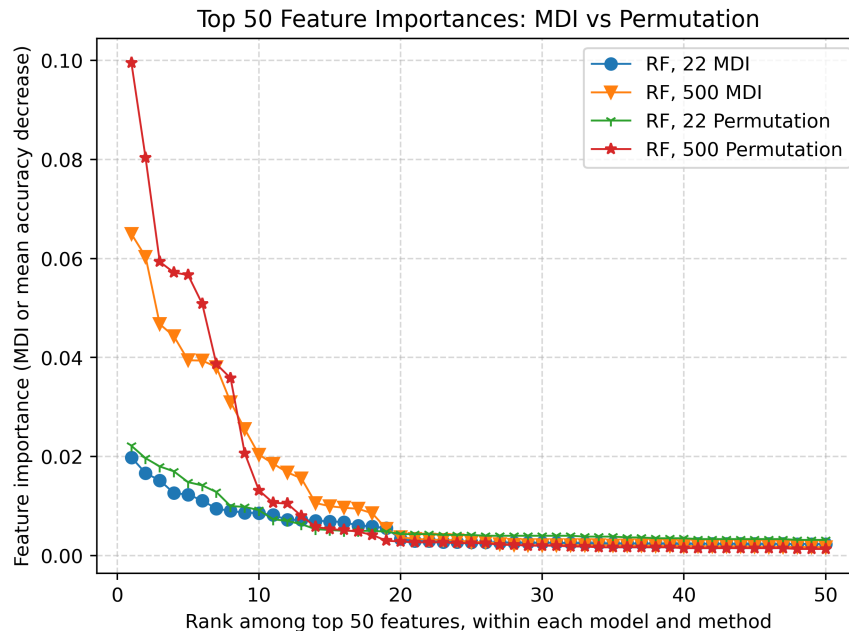


Figure 1: Feature Importances for the two RF, computed via MDI and via Permutations, part b.

- (1c) Selecting k most important features.

Label	k	Train Misclass. Error	Test Misclass. Error
RF_22_MDI	10	0.000000	0.153333
RF_22_MDI	20	0.000000	0.116667
RF_22_MDI	30	0.000000	0.126667
RF_22_MDI	40	0.000000	0.131667
RF_500_MDI	10	0.000000	0.116667
RF_500_MDI	20	0.000000	0.115000
RF_500_MDI	30	0.000000	0.131667
RF_500_MDI	40	0.000000	0.136667
RF_22_PERM	10	0.000000	0.131667
RF_22_PERM	20	0.000000	0.131667
RF_22_PERM	30	0.000000	0.146667
RF_22_PERM	40	0.000000	0.130000
RF_500_PERM	10	0.000000	0.128333
RF_500_PERM	20	0.000000	0.118333
RF_500_PERM	30	0.000000	0.136667
RF_500_PERM	40	0.000000	0.138333

Table 2: Training and test misclassification errors for the fitted models, part c.

(1d) asdf

(1e) Of the models,

Code